# SIMATS ENGINEERING
## ASSESSMENT TOOL 1
## Experiments

**COURSE NAME: Computer Networks** **COURSE CODE: CSA07**

**COURSE OUTCOME COVERED**

**CO3:** *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

***Code of Conduct:***

*I* G. Santosh Kumar *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student***
G. Santosh Kumar

36/40

## Experiments

**Total Marks:40**

**QUESTIONS:4**

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25 | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design | 30 | Well-planned, systematic implementation with optimal solution | Correct implementation with minor issues | Basic implementation with errors or inefficiencies | Incorrect or incomplete implementation |
| Use of Tools | 20 | Effective and advanced use of tools/technologies with proper configuration | Appropriate use of tools with correct execution | Limited or inconsistent use of tools | Incorrect or minimal use of tools |
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable interpretation | Basic analysis with limited insights | Poor or incorrect analysis of results |
| Documentation | 10 | Clear, well-structured documentation with diagrams, code, and results | Organized documentation with necessary details | Basic documentation with missing details | Poor or incomplete documentation |

TCP three-way Handshake

The TCP connection is established using a three-way handshake
The three packets are :

1. SYN
2. SYN-Ack
3. Ack.

1. SYN packet

The client sends a packet with the SYN flag set to 1.

Ex :-

client → server: SYN, sequence Number = 1000

The SYN packet request a TCP connection & provides the client's initial sequence number.

2. SYN-Ack packet

The server responds with both SYN & Ack flags.

server → client : SYN+Ack, sequence Number = 5000,

Ack = 1001

• SYN indicates that the server excepts the next sequence number from the client to be 1001.

3. Ack packet

The client sends the final acknowledgement.

client → server : Ack, Ack = 5001

Role of the flags

| packet | flag | purpose |
|---|---|---|
| 1. | SYN | Starts connection & synchronizes sequence numbers |
| 2. | SYN+ACK | Accepts request & client |
| 3. | ACK | confirms server response. |

Wireshark observation

In wireshark, use:

tcp.flags.syn==1

SYN → SYN/Ack → Ack.

Q. DNS using UDP & comparison with TCP

DNS commonly uses UDP for normal queries because UDP has a smaller & simpler header.

UDP header size

The UDP header is only 8 bytes.

Its fields are:

- Source port – 2 bytes
- Destination port – 2 bytes
- Length – 2 bytes
- Check sum – 2 bytes

Note:

UDP header = 8 bytes

TCP header size

A normal TCP header is at least 20 bytes. So it can become larger when TCP options are present.

DNS over UDP

A typical DNS query looks like:

client → DNS server : UDP query

The server responds:

DNS server → client : UDP response

DNS traditionally uses port 53 for UDP.

Comparison

Why DNS uses UDP

DNS queries are generally small & require low communication overhead. UDP follows a client DNS to send a query without first establishing a connection.

Wireshark:-

A useful filter is:

dns

or:

udp.port == 53

In wireshark, the DNS packet can be inspected to see the Query, response, transaction ID & requested domain.

3. TCP Retransmission vs UDP During packet loss

TCP & UDP handle packet loss differently.

TCP:—

TCP provides reliable delivery. when a packet is lost, TCP detects the loss using acknowledgement & sequence numbers.

Example:—

packet-1 → received
packet-2 → lost
packet-3 → received

The receiver can indicate that it is still expecting the missing data. TCP can then retransmit the lost segment

TCP segment 2 → lost
TCP detects loss → Retransmission
TCP segment 2 → received.

UDP:—

..., it does not provide built-in retransmission.

If a UDP packet is lost:

UDP packet → Lost

UDP simply continuous without automatically sending the packet again.

Wireshark Observation

TCP retransmission can be identified using:

tcp.analysis.retransmission

Wireshark may display:

TCP retransmission

Why TCP is more reliable

TCP uses:-

- Sequence numbers.
- Acknowledgement.
- Retransmission.
- flow control.
- Congestion control.
- Ordered delivery.

UDP does not provide these mechanisms.

TCP:-

send → wait/Ack → Detect loss → Retransmit

UDP:-

send → continue

4. DNS traffic Capture & Query types

Ans:— DNS converts human-readable domain names into IP address.

Eg:—

www.example.com → IP address.

wireshark can be used to capture & analyze this DNS communication.

Step 1:- Start wireshark

Start wireshark & select the active network interface such as:

- wi-fi
- ethernet

St:-02 Apply DNS filter

use:

dns

This displays DNS packets.

you can also use:

udp.port==53

to display DNS traffic over UDP port 53

St:-03 Resolve a domain

for ex, open a browser & access a domain such as:

example.com

Computer sends a DNS query to a DNS browser.

The communication can be represent as:

Client → DNS server : Query

DNS server → Client : Response

DNS Query types

A Record

The A record requests an IPV4 address.

Ex:-

Example.com → 93.184.216.34

AAAA Record

The AAAA record requests an IPV6 address.

Ex:-

Example.com → IPV6 address.

DNS resolution Time

wireshark provides timestamps for packets.

Ex:-

DNS query : 0.000000 s

DNS response = 0.025000 s

Therefore =

Resolution time = Response time - Query time

= 0.025000 - 0.000000

= 0.025 seconds.